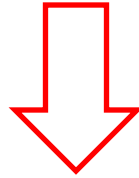


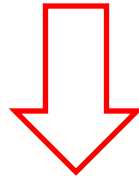
Chapter 7

The labor market

The medium run



Firms adjust prices and wages



These changes affect output

Labor market at the center of that sequence of events

Start with the labor market

Remember from first lecture:

1. **Working age population:** People between 15-65 years old
2. **Labor force:** Unemployed + Employed
3. **Unemployed:** Do not have a job, but **actively look for** one
4. **Out of the labor force:** Not working and not looking for work
5. **Discouraged workers:** People without jobs who give up looking for work but would like one

Participation rate

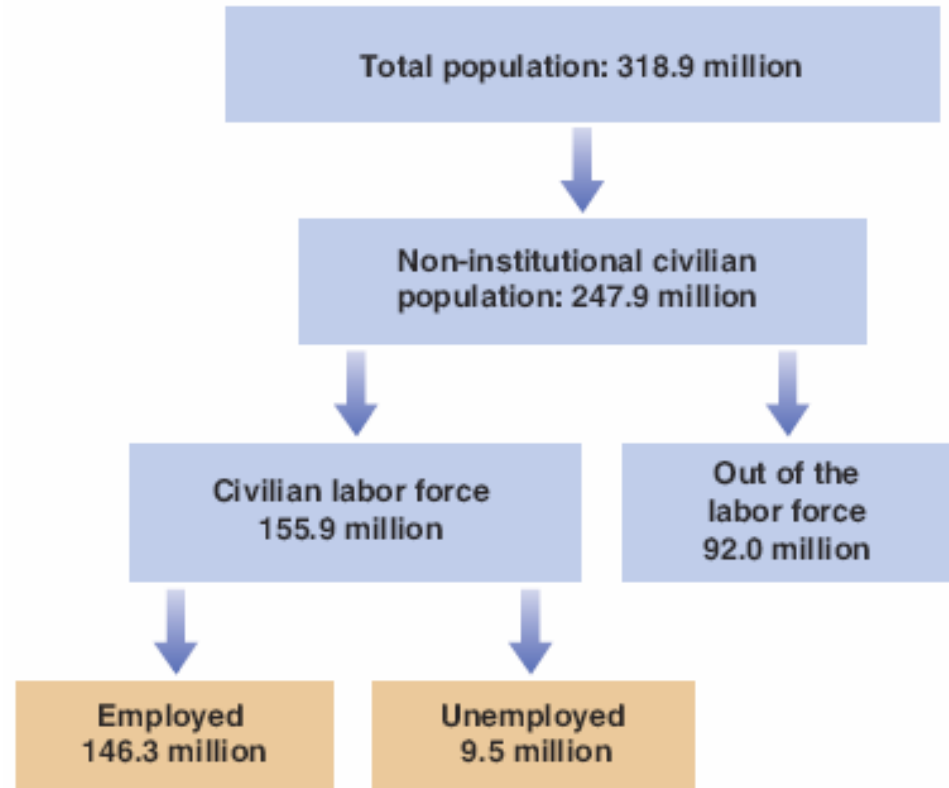
$$\text{Participation rate} = \frac{\text{Labor force}}{\text{Population of working age}}$$

Unemployment rate

$$\text{Unemployment rate} = \frac{\text{Unemployed}}{\text{Labor force}}$$

Population, labor force, employment and unemployment in the United States, 2014

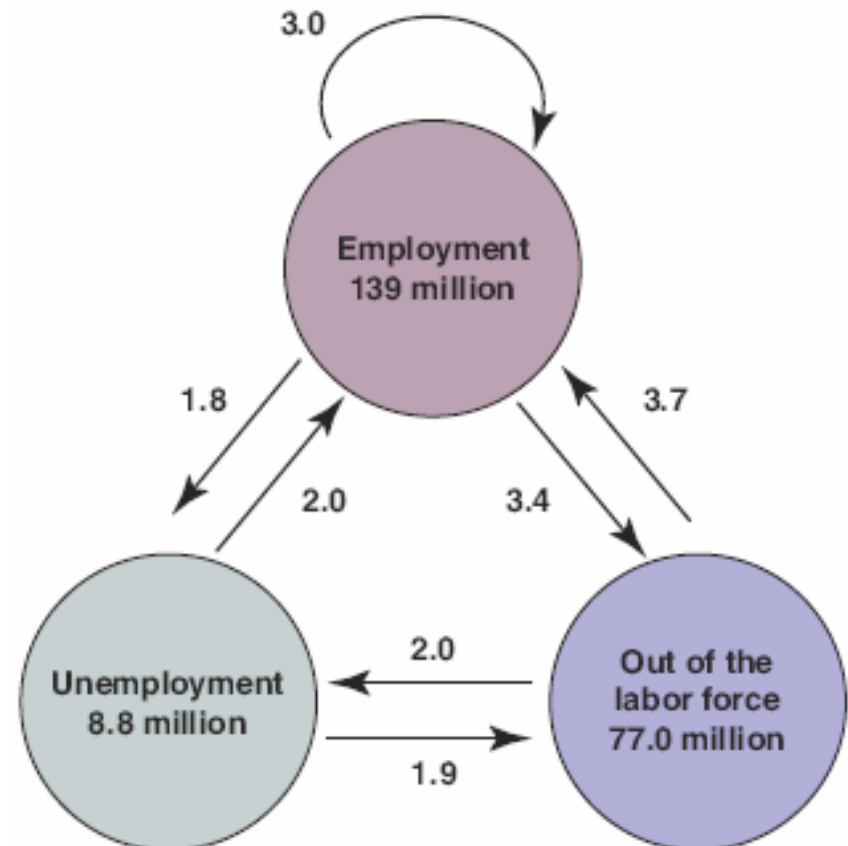
- The unemployment rate is the ratio of the unemployed to the labor force, it was $9.5/155.9 = 6.1\%$
- The participation rate is the ratio of the labor force to the population, it was $155.9/247.9 = 62.8\%$



- The labor market is in general very dynamic!
- In general, we observe large **flows** of people between employment, unemployment, and non-participation
- However, a given unemployment rate may reflect
 - A **dynamic labor market**, with **many separations** (workers who are leaving for another job and workers losing their jobs) and **many hires** (workers newly employed by firms)
 - A **sclerotic labor market** with **few separations**, **few hires** and a stagnant unemployment pool

Average monthly flows between employment, unemployment, and non-participation in the United States, 1996-2014 (millions)

- The flows of workers in and out of employment are large
- The flows in and out of unemployment are large relative to the number of unemployed
- There are also large flows in and out of the labor force, much of it directly to and from employment



The Current Population Survey (CPS) measures the average monthly flows

- Large **flows** of workers **in and out of employment** (hires and separations)

Note that **separations** consist of

- **Quits** (workers leaving their jobs for a better alternative)
 - **Layoffs** (which come from changes in employment levels across firms)
-
- Large **flows in and out of unemployment**

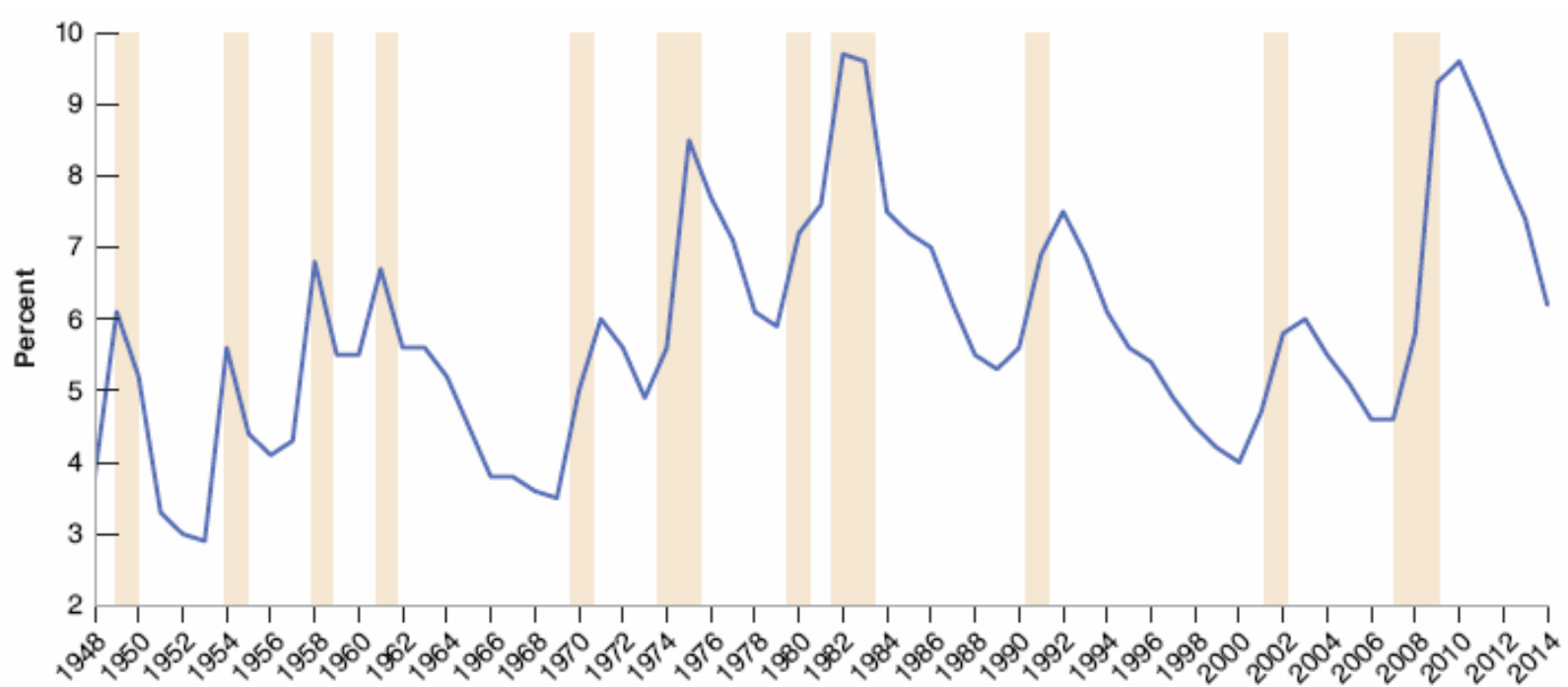
Average **duration of unemployment** in US is between 2 and 3 months (dynamic)

Average **duration of unemployment** in Europe is much longer (sclerotic)

- Large **flows in and out of the labor force**, much of them directly to and from employment

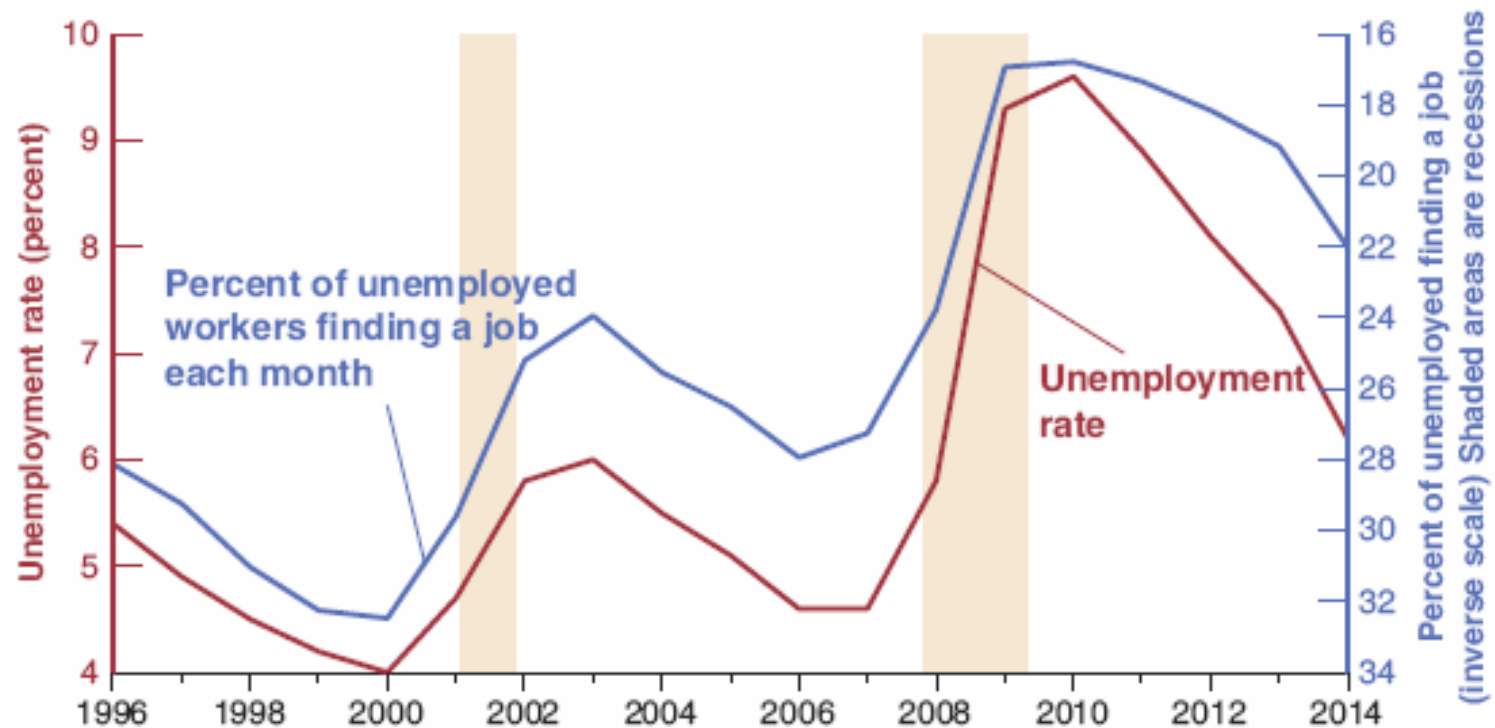
In particular, **discouraged workers** are classified as “out of the labor force”, but they may take a job if they find it

Unemployment rate in the United States, 1948-2014



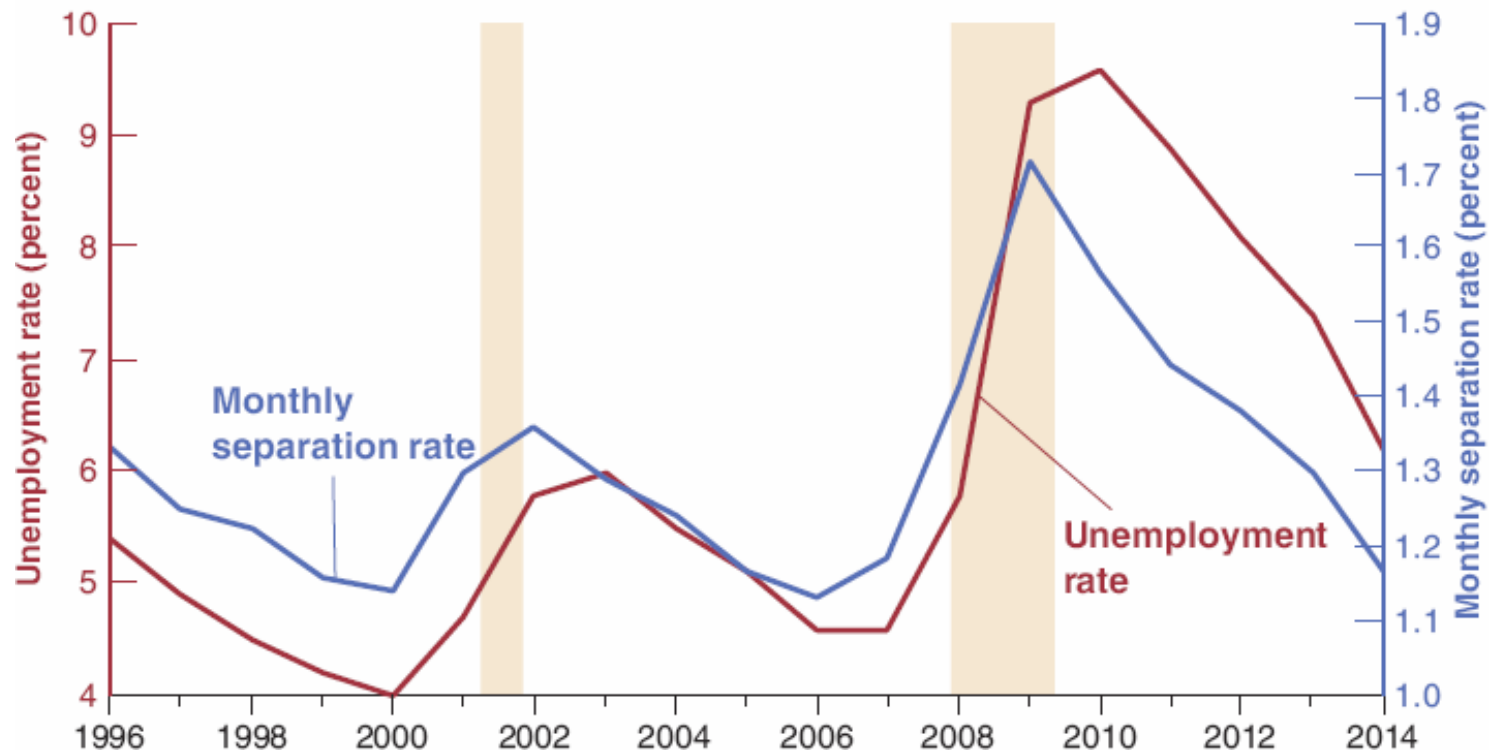
Since 1948, the average yearly U.S. unemployment rate has fluctuated between 3 and 10%

Unemployment rate and proportion of unemployed finding jobs in the United States, 1996-2014



When unemployment is higher, the proportion of unemployed finding jobs within one month is lower. Note that the scale on the right is an inverse scale

Unemployment rate and separation rate in the United States, 1996-2014



When unemployment is higher, a higher proportion of workers lose their jobs

When unemployment is high, workers are worse off in two ways:

- Employed workers face a higher probability of losing their job
- Unemployed workers face a lower probability of finding a job; or they can expect to remain unemployed for a longer time

Wage determination

How do firms and workers set wages?

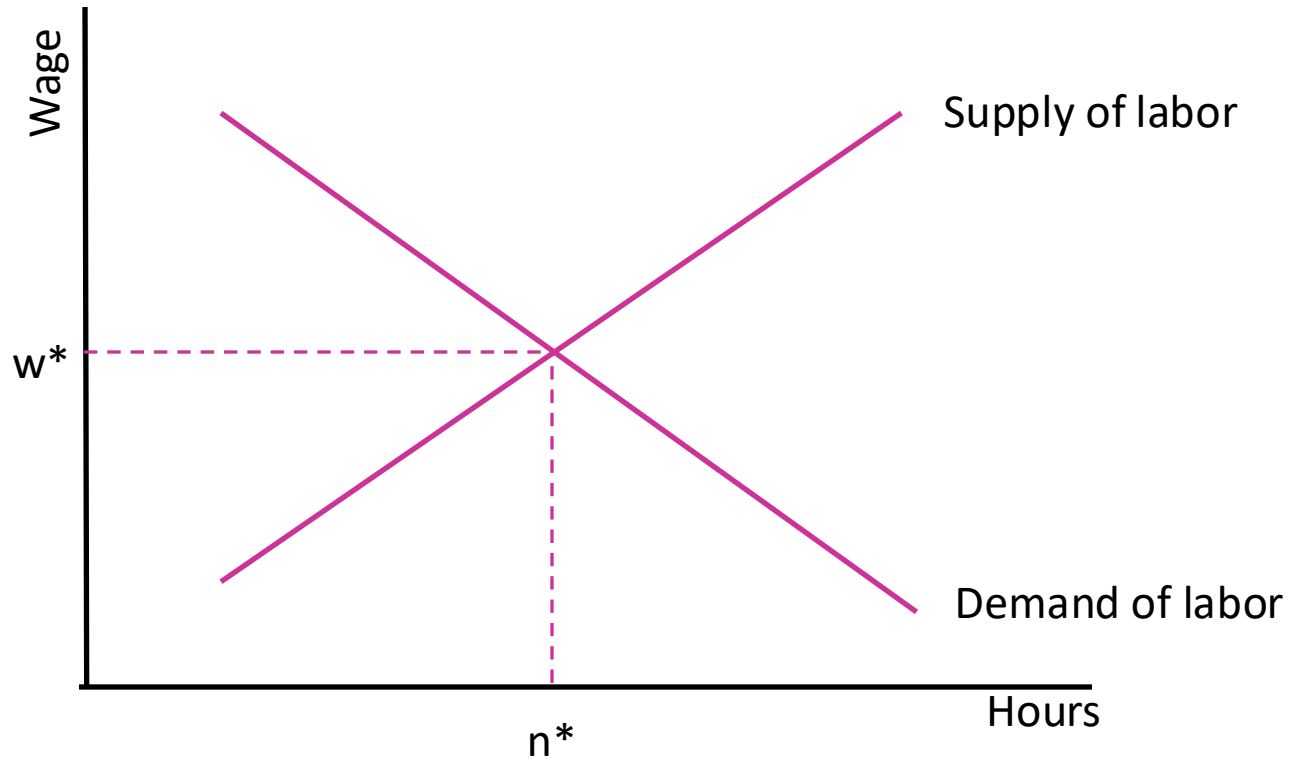
- **Collective bargaining** (unions...)
High in Europe (about 90% of workers covered by collective bargaining in Germany and France), low in USA (slightly more than 10% of workers)

Alternatively

- **Take-it-or-leave-it offers**
Especially for low-skilled workers
- **Workers negotiate directly with firms**
Especially for high-skilled workers

What is wrong with what you learned in Microeconomics?

**Where is
unemployment?**



In general, there are **common forces** at work

1. Workers are typically paid a wage higher than their **reservation wage**, the wage that makes them indifferent between working or being unemployed
2. Wages typically depend on **labor market conditions**. The lower the unemployment rate, the higher the wages

How can we **explain these facts?** Two theories:

1. Bargaining
2. Efficiency wages

1. Bargaining

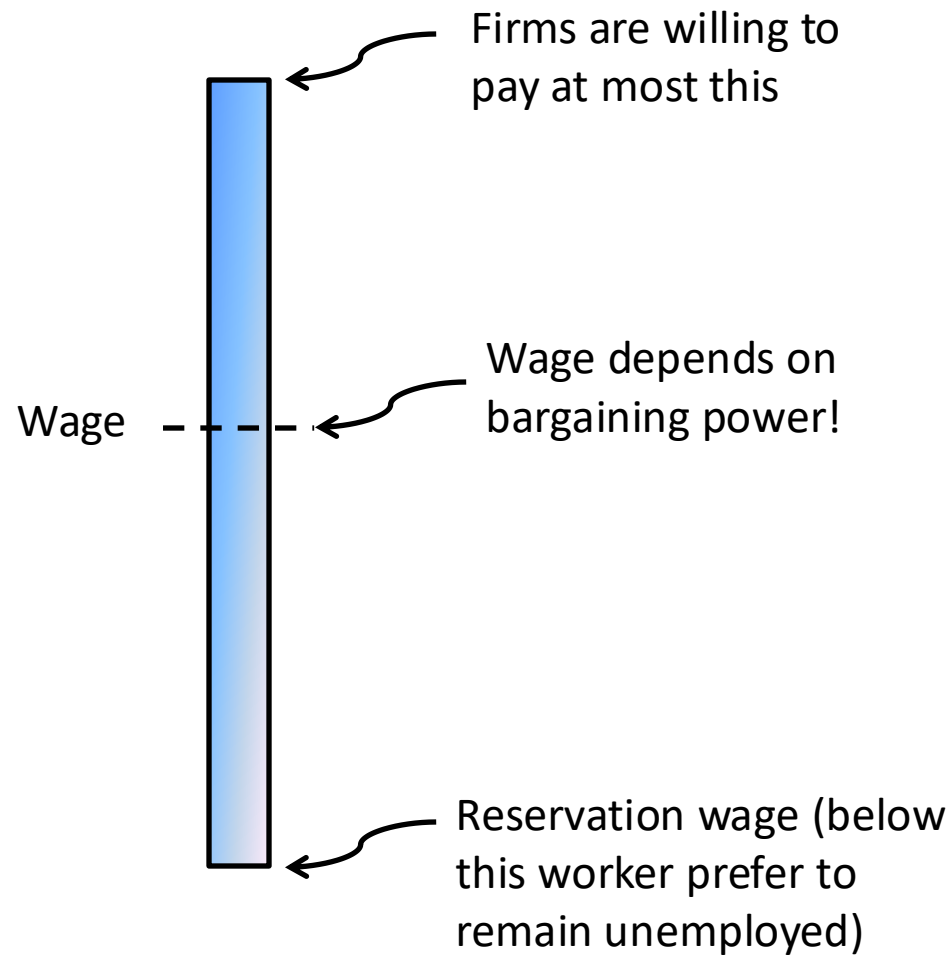
How much **bargaining power** a worker has depends on two factors

- How costly it would be for the firm to replace him
- How hard it would be for him to find another job

Bargaining power will then depend on

- **Skills**: highly skilled workers are more difficult to replace; they will have more bargaining power
- **Labor market conditions**: when the unemployment rate is low, it is easier for workers to find other jobs and more difficult for firms to find replacement workers; workers will have more bargaining power

Wage bargaining set



2. Efficiency wages

Firms may want to pay higher wage than the reservation wage to **incentivize workers** to be productive (efficient) or to stay at the firm

- **Nature of the job**: firms (such as high-tech firms) that see employee morale and commitment as essential will pay more than firms in sectors where workers' activities are more routine
- **Labor market conditions**: when unemployment is low, quits and turnover will be higher, and firms will pay higher wages to compensate and retain workers

Annual turnover and layoff rates (%) at Ford, 1913–1915			
	1913	1914	1915
Turnover rate	370	54	16
Layoff rate	62	7	0.1

- Turnover rate: ratio of separations to employment
- Layoff rate: ratio of layoffs to employment
- In 1914, Henry Ford, the builder of Model-T, announced that his company would pay all qualified employees a minimum of \$5.00 a day for an eight-hour day, compared to previously an average \$2.30 for a nine-hour day
- The turnover rate plunged from 370% in 1913 to 16% in 1915
- The layoff rate collapsed from 62% to nearly 0%

Many different theories, but they can all be represented as

$$W = P^e F(u, z)$$

(−, +)

Nominal wage, W , depends on 3 factors

- the **expected price level**, P^e
- the **unemployment rate**, u
- a catchall variable, z , representing **all other variables** that may affect the outcome of wage setting

Let's see each of them in turn.

The expected price level

First ignore the difference between P and P^e

Both workers and firms care about **real wages** (W/P), **not nominal wages** (W)

- Workers do not care about how many dollars they receive but about **how many goods they can buy** with those dollars. They care about W/P
- Firms do not care about the nominal wages they pay but about the nominal wages, W , they pay **relative to the price of the goods they sell**, P . They also care about W/P

Now, **why** do **wages depend on the expected price level P^e** as opposed to the actual price level P ?

Because **wages are set in nominal terms**, and when they are set the price level is not yet known

- Think about union contracts setting nominal wages in advance for one or more years
- Even when wages are set by firms or by bargaining between workers and firms, nominal wages are typically set for one year and if the price level raises unexpectedly during the year, nominal wages are not readjusted

The unemployment rate

- If we think of wages as being determined by bargaining, then higher unemployment weakens workers' bargaining power, forcing them to accept lower wages
- If we think of wages as being determined by efficiency wages considerations, then higher unemployment allows firms to pay lower wages and still retain workers

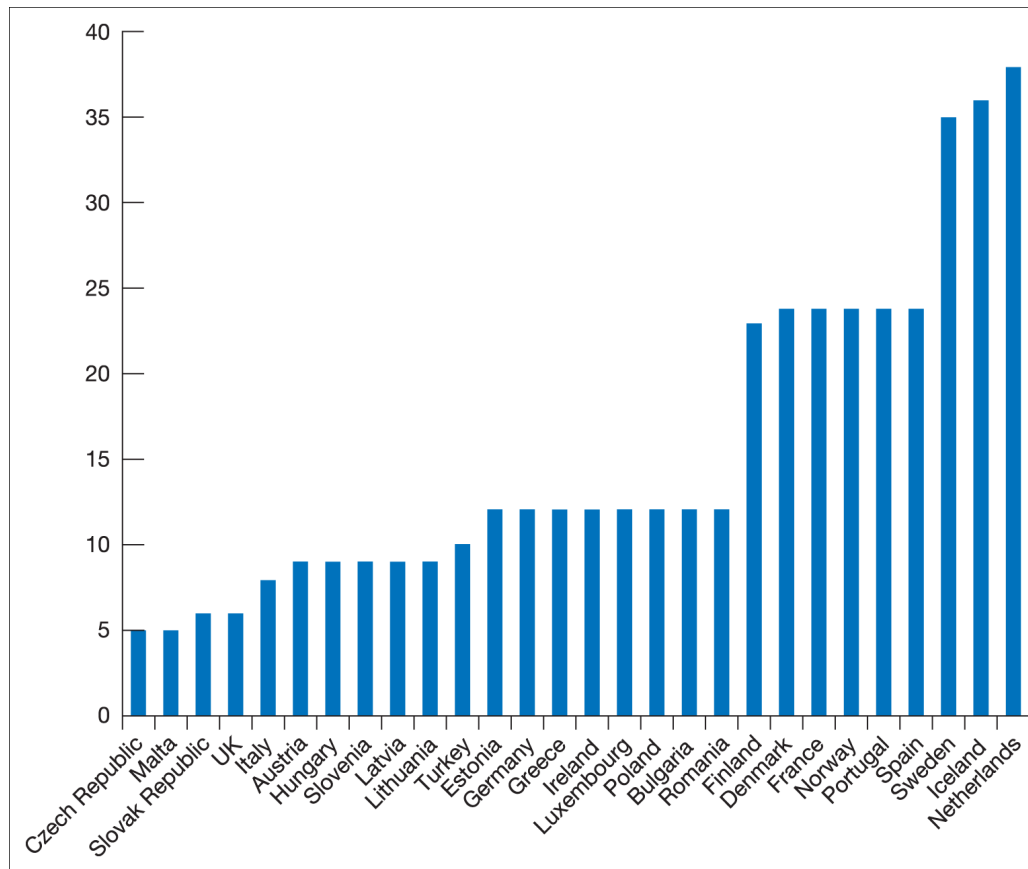
The other factors

The third variable, z , is a catchall variable that stands for all the factors that affect wages other than P and u

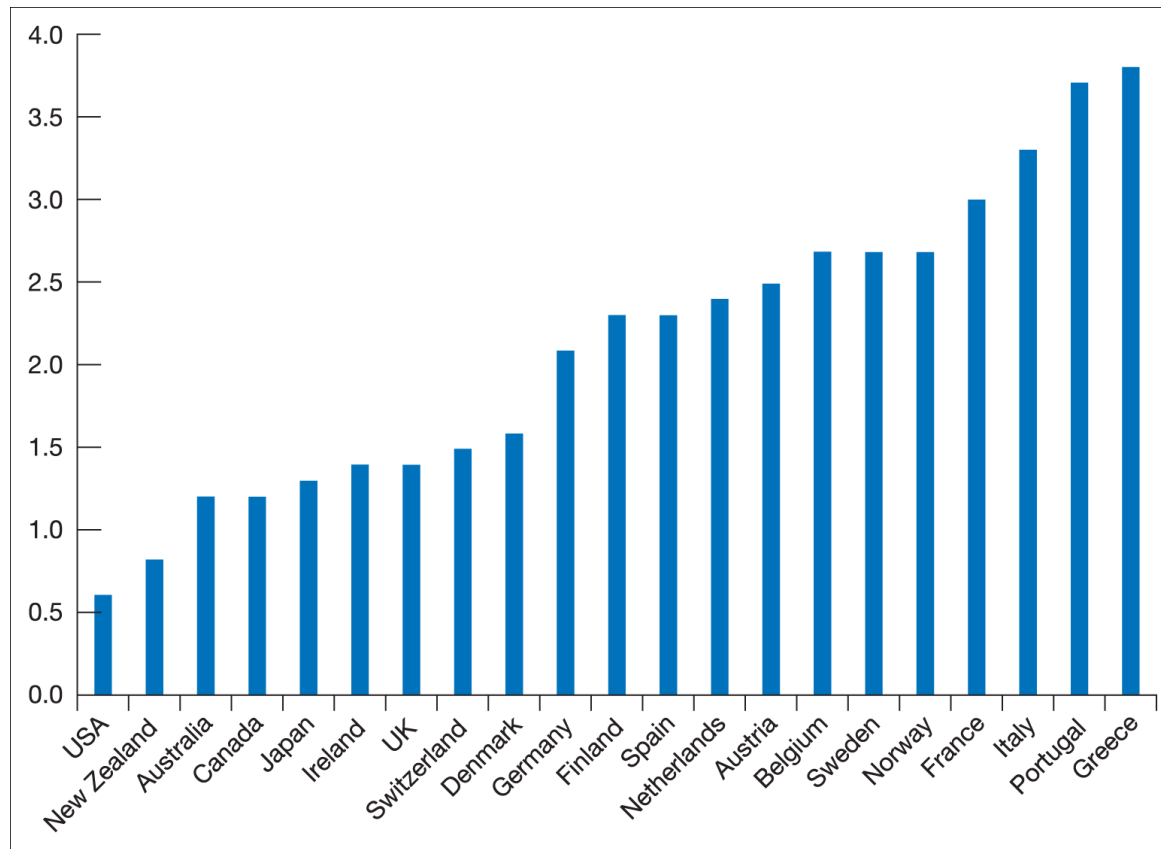
Examples:

1. **Unemployment insurance:** Payment of unemployment benefits to workers who lose their jobs
2. **Employment protection:** How easily can firms layoff workers?
3. **Minimum wages**

Duration of unemployment insurance in Europe, 2010 (in months)



Employment protection in Europe, 2003 (index)



Price determination

- **Production function**: quantity of **output Y** produced as a function of the inputs used in production
- Assuming that firms produce using only **labor N**, gives

$$Y = AN$$

where **A is labor productivity** (output per worker)

Assume **for simplicity** that one worker produces one unit of output: **A=1**

$$Y = N$$

How do firms set prices?

- Perfect competition
- Imperfect competition (markup)

- We assume **imperfect competition**, firms set their price according to:

$$P = (1 + m)W$$

where

- $m > 0$ is the **markup**
 - W is the **marginal cost** (the cost of producing one additional unit of output)
- Under perfect competition, prices would be given by

$$P = W$$

where the markup $m = 0$

- *Let's work out the model also with A not equal 1!*
- Firms set their price according to:

$$P = (1 + m) \frac{W}{A}$$

where

- $m > 0$ is the **markup**
- W/A is the **marginal cost** (the cost of producing one additional unit of output)

The natural rate of unemployment

- Back to unemployment...
- Assume for now that nominal wages depend on the **actual price level**, P , rather than on the expected price level, P^e
- Wage setting and price setting relations determine the **equilibrium rate of unemployment**

Remember **wage-setting**:

$$W = P^e F(u, z)$$

Now:

$$W = PF(u, z)$$



$$\frac{W}{P} = F(u, z)$$

(-,+)

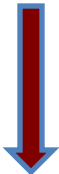
Real wage

- This negative relation between the real wage and the unemployment rate is called the **WAGE-SETTING RELATION**
- The wage setting relation gives the **real wage targeted by workers**

Remember **price-setting**:

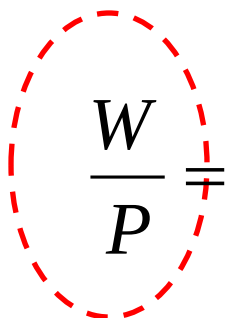
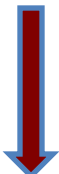
$$P = (1 + m)W$$

If we divide both sides by W :


$$\frac{P}{W} = (1 + m)$$

Or equivalently:

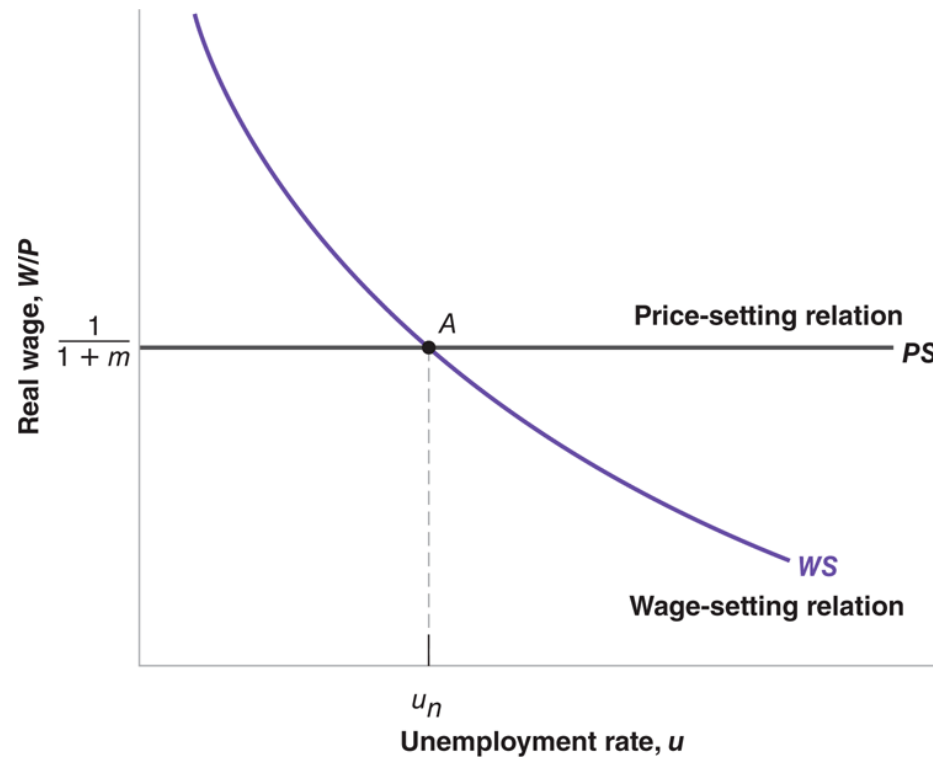
Real wage


$$\frac{W}{P} = \frac{1}{1 + m}$$

This is called the **PRICE-SETTING RELATION**

The price setting relation gives the **real wage that firms are willing to pay**

Wage and price-setting relations



- In equilibrium: **real wage targeted by workers** (from **wage-setting relation**) equals **real wage firms are willing to pay** (from **price-setting relation**)
- This determines the equilibrium unemployment rate
- The equilibrium unemployment rate (u_n) is called the **natural rate of unemployment**

with math...

$$\frac{W}{P} = F(u, z)$$

wage-setting relation

$$\frac{W}{P} = \frac{1}{1+m}$$

price-setting relation



$$F(u_n, z) = \frac{1}{1+m}$$

The **natural rate of unemployment** (u_n) will depend on factors like unemployment benefits (z) and product market regulation (m)

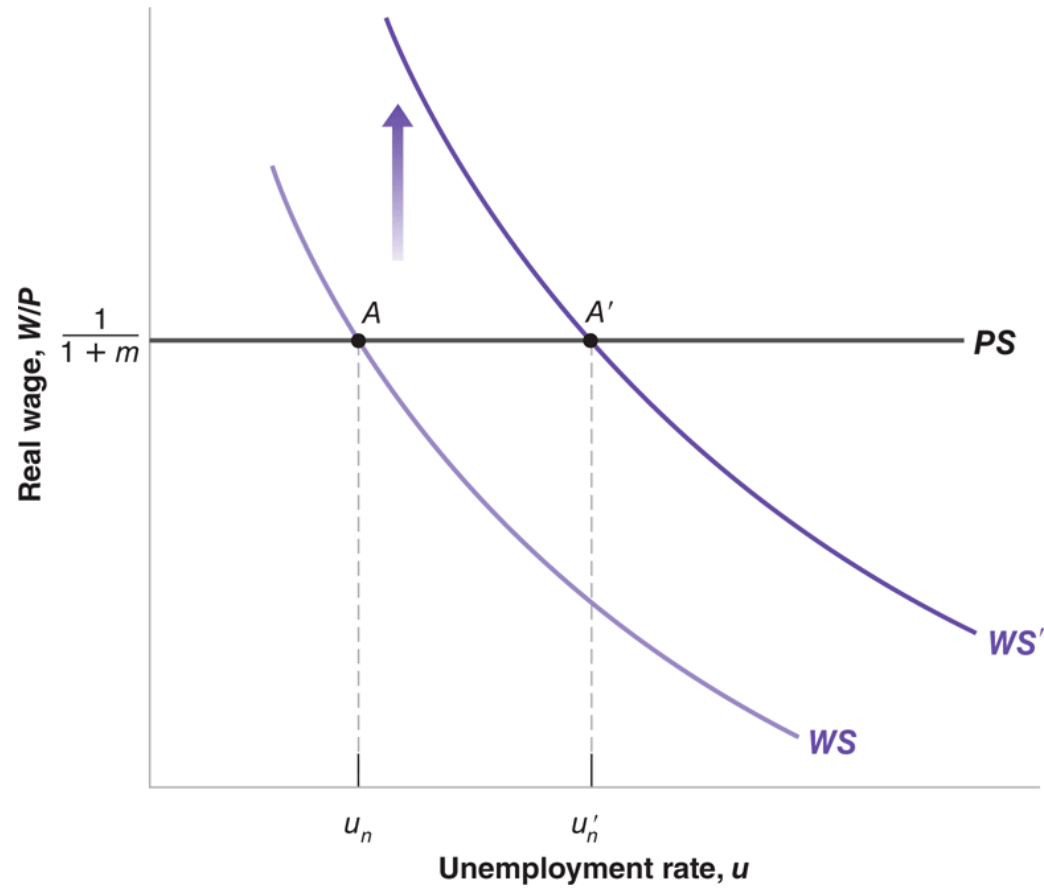
Effects of an increase in z

- Example: **higher unemployment benefits**
- Graphical analysis: A raise in z shifts the wage-setting relation up

$$\frac{W}{P} = F(u, z)$$

- Effect: The unemployment rate increases and wages remain constant
- Economic intuition: workers demand higher real wages; the wage firms are willing to pay is unchanged; unemployment rate increases to restore equilibrium

z increases



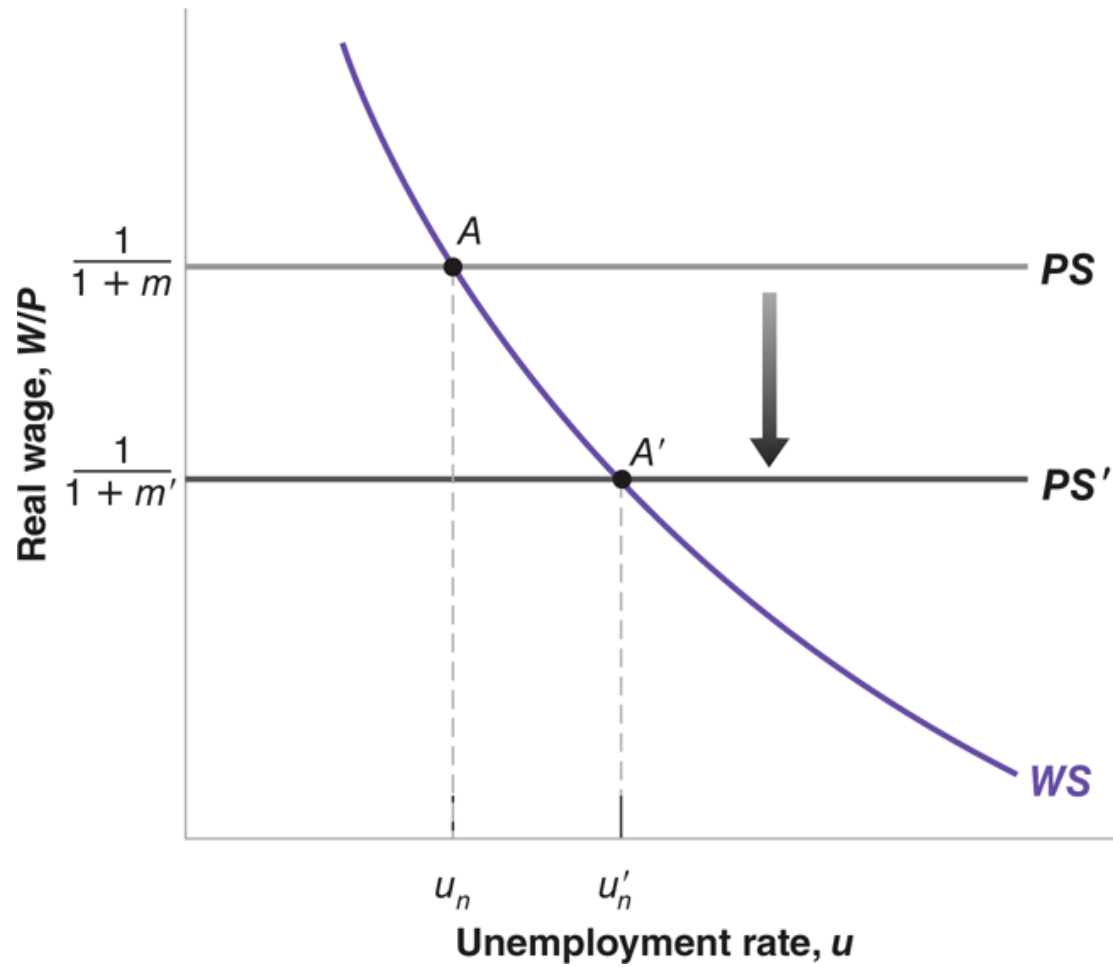
Effects of an increase in m

- Example: **less stringent competition law** (firms collude and charge higher markups)
- Graphical analysis: A raise in m shifts the price-setting relation down

$$\frac{W}{P} = \frac{1}{1+m}$$

- Effect: The unemployment rate increases and wages decrease
- Economic intuition: real wage paid by firms decreases; unemployment rate increases so that people accept lower wages

m increases



So far only unemployment, how about **employment**?

$$u = \frac{U}{L} = \frac{L - N}{L} = 1 - \frac{N}{L}$$

Therefore, rearranging

$$N = L(1 - u)$$

The **natural level of employment**, N_n , is given by:

$$N_n = L(1 - u_n)$$

Associated with the natural level of employment is the **natural level of output**, and ...

Since $Y=N$:

$$Y_n = N_n = L(1 - u_n)$$

The natural level of output satisfies the following:

$$F\left(1 - \frac{Y_n}{L}, z\right) = \frac{1}{1 + m}$$

Short and medium run

- We have focused on the labor market, and we have assumed that the price level is equal to the expected price level
- In the **short run**, the **price level** may well turn out to be **different from what is expected** when nominal wages are set, so that unemployment is not necessarily equal to the natural rate or output equal to its natural level
- In the short run output continues to be determined by factors such as fiscal and monetary policy
- Because **expectations are unlikely to be systematically wrong**, in the **medium run**, output tends to return to its natural level
- The next chapter will relax the assumption that the price level is equal to the expected price level and derive a relation between unemployment and inflation